modus-wc-time-input — custom time picker

Modus Web Components 2.0 · implementation notes · 6 November 2026

What changed

The time field is still a native `<input type="time">`. The clock button no longer opens the browser popup. It opens a custom hour and minute wheel. The stored value stays a 24-hour `HH:mm` or `HH:mm:ss` string.

The wheel layout comes from pull request 1430 (`1216-time-picker-component`). The utilities live under `src/components/modus-wc-time-input/utils/`.

How it is made

The field and the clock sit in one `<label>`. Chromium's clock glyph is removed with CSS. Firefox does not allow that, so its glyph is slid under the custom button and covered. Pressing the custom clock does not call `showPicker()`. It toggles `TimePickerDropdown`. Popper places that panel under the input. A wheel click writes the new time back onto the same input and emits `inputChange`.

Implementation steps

1. Keep the native input

`type="time"` still owns `value`, `min`, `max`, `step`, `name`, and form submit. The custom panel only edits that value. See the `<input>` in `modus-wc-time-input.tsx` (around the render method).

2. Replace the browser clock with modus-wc-button

The button is hidden when the field is disabled or read-only (`shouldShowTimePickerButton`). Its size follows the input size: a `sm` input uses an `xs` icon (`getTimePickerToggleSize`).

3. Stop the label from opening the browser picker

The button is inside the `<label>`. A custom element is not interactive content, so a click would activate the time input and open the native picker. `mousedown` calls `preventDefault()` before that happens, then toggles the panel.

```
private handleTimePickerPointerDown = (event: MouseEvent) => {
  event.preventDefault();
  event.stopPropagation();
  this.showDropdown = !this.showDropdown;
  this.pendingScrollToSelection = this.showDropdown;
};
```

`onButtonClick` receives a Stencil `CustomEvent<MouseEvent | KeyboardEvent>`, not a raw `MouseEvent`. That handler only stops the event. The panel opens on `mousedown`, because cancelling `mousedown` also cancels the

later click in some browsers.

4. Block the Chromium keyboard shortcut

Alt+ArrowDown and Alt+ArrowUp normally open the native picker. `handleInputKeyDown` cancels those keys and opens the custom panel instead.

5. Render the wheel panel from PR 1430

- `utils/time-options.ts` and `utils/time-format.ts` build the columns and convert wheel state to a 24-hour string.
- `utils/time-picker-dropdown.tsx` draws the hour, minute, and optional second columns.
- `utils/time-wheel-scroll.ts` scrolls to the selection and loops long columns.

The format passed in is fixed at `24hrs`. A minute column uses `step` only when `step` is a whole number of minutes that divides 60 (`getMinuteStep`). Seconds appear when `showSeconds` is set or `step` is under 60 seconds (`shouldShowSeconds`).

6. Position the panel

`componentDidUpdate` creates a Popper instance anchored to the input (`createPopperOptions('bottom-start', 4)`) and calls `scrollWheelsToSelection`. Escape, a click outside the host, or committing a row closes the panel. `disconnectedCallback` destroys Popper and the scroll listeners.

7. Write the selection back through the input

```
const next24h = valueFromWheelState({ ...current, ...partial }, showSeconds, '24hrs');
this.value = next24h;
this.timeInputRef.value = next24h;
this.timeInputRef.dispatchEvent(new Event('input', { bubbles: true }));
```

Existing `inputChange` listeners still see a native `input` event, because `handleInput` emits that event.

8. Hide the extra browser clock

Chromium's indicator is `display: none` on `::-webkit-calendar-picker-indicator` in `modus-wc-time-input.scss`. Firefox ignores that pseudo-element. There the input is pulled `2rem` under the custom button, and the button container paints `--modus-wc-color-base-page` over the native glyph. That rule is limited to Firefox with `@supports (-moz-appearance: textfield)`.

Advantages

- The popup chrome is the same Modus wheel in every browser, instead of each browser's own time panel.
- The public value stays a 24-hour time string on a real time input, so forms, `min` / `max`, and current `inputChange` handlers keep working.

- The wheel layout, keyboard rows, and circular scroll already exist on the PR 1430 branch.
- Disabled and read-only fields do not show the custom clock.

Disadvantages

- Firefox cannot delete its clock. Covering it is a layout overlap. If the button is narrower than the native glyph, or the field background is not `--modus-wc-color-base-page`, the gray clock can show at the edge.
- The native picker is not fully gone. Clicking the time text, or a browser shortcut other than Alt+Arrow, can still open it, because the control is still `type="time"`.
- The panel is 24-hour only. PR 1430's `12hrs` format and AM/PM column are not wired.
- `step` values that are not a clean minute interval fall back to every minute, so the wheels can offer times the native `step` would reject.
- Choosing a time updates the input by setting `.value` and dispatching a plain `input` event. That is a second write beside Stencil's controlled `value` prop.